P1 , P2 Δf(resoluccio f.)= Fs/Ntotal(mostes + zero pading)
Fs = frecuencia de mosteig(mostes/s) T(periode)=1/f
dim = dimensions(mostres) → punts que retornarà l'arrey
Tobs=(s)
dim (mostres)=Tobs*Fs (mostres/s * s) factor = dim nova/dim

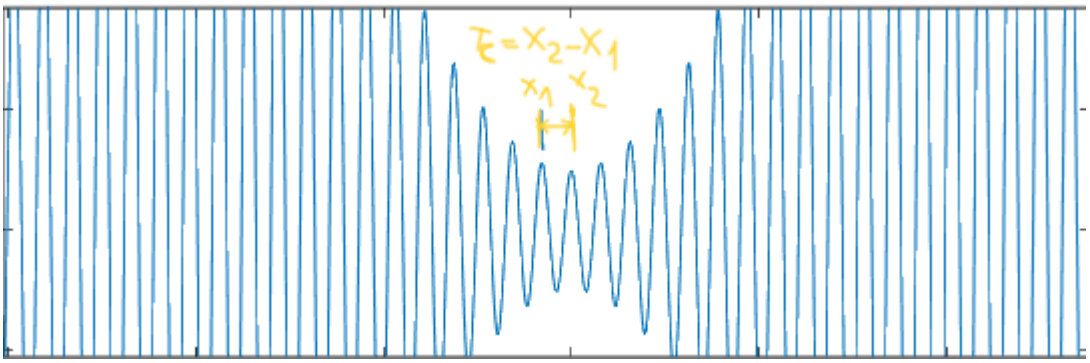
zero padding?

nova dim = Fs/resolucio de freq.
num zeros = nova dim-dim inicial(enunciat)

P3

m= Emax-Emin/Emax+Emin → en senyal de temps (com la Tm)
Emax → Y mes alta
Emin → Y mes baixa

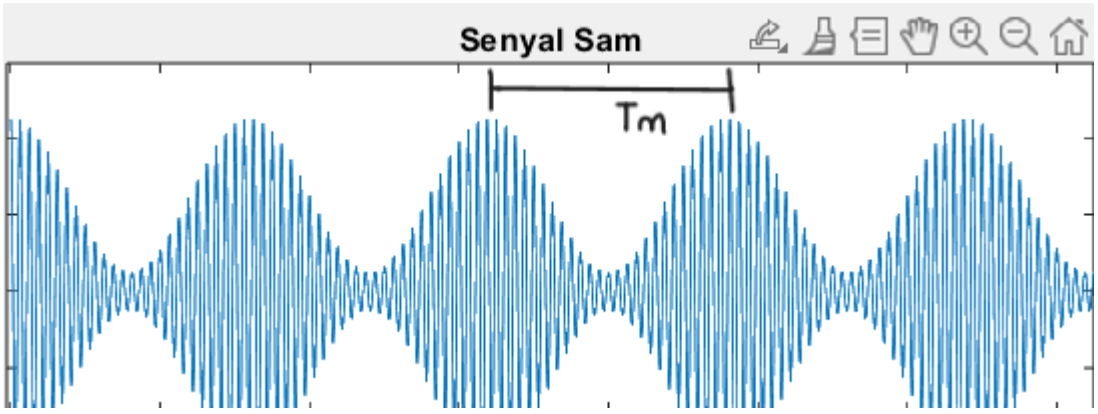
Tc= diferencia entre dos punts consecutius de X



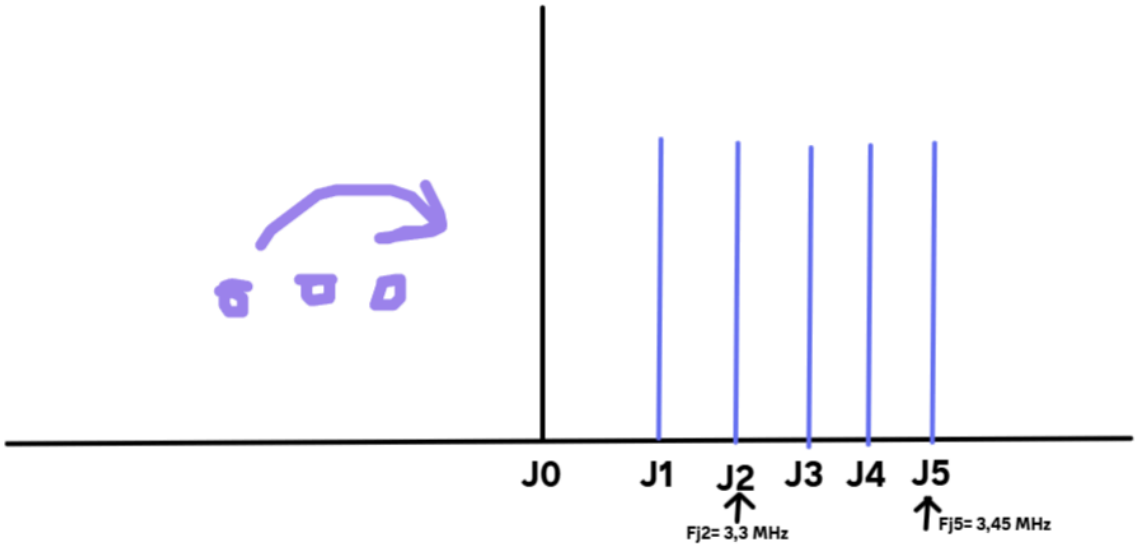
$T_c = 1{,}898 \cdot 10^{-5} - 1{,}699 \cdot 10^{-5} = 0{,}199 \cdot 10^{-5}\ s$

Fc=1/Tc

Tm/2 = x (cuan y min) - x (cuan y max)
Tm= x (cuan y max) - x (cuan y torna a ser max) → Fm=1/Tm



P4



Fc ←→Fm

Fjn=Fc+-n*Fm

Fj2= Fc + 2*Fm = 3,3 MHz
Fj5= Fc + 5*Fm = 3,45 MHz

3,45-3,3 = 3Fm
Fm = 0,15/3 MHz
Fc = Fj2 - 2*Fm = 3,3 Mhz - 2*0,15/3 Mhz = 3,2 Mhz

Beta → mirar a la taula com hi ha fins a J5 llavors → beta = 2 J0= 0,22 (eix Y)

Funciones de Bessel de primera clase $J(\beta)$

Índice de modulación	Portadora	Pares laterales de frecuencia													
β	J_0	J_1	J_2	J_3	J_4	J_5	J_6	J_7	J_8	J_9	J_{10}	J_{11}	J_{12}	J_{13}	J_{14}
0.00	1.00	—	—	—	—	—	—	—	—	—	—	—	—	—	—
0.25	0.98	0.12	—	—	—	—	—	—	—	—	—	—	—	—	—
0.5	0.94	0.24	0.03	—	—	—	—	—	—	—	—	—	—	—	—
1.0	0.77	0.44	0.11	0.02	—	—	—	—	—	—	—	—	—	—	—
1.5	0.51	0.56	0.23	0.06	0.01	—	—	—	—	—	—	—	—	—	—
2.0	0.22	0.58	0.35	0.13	0.03	—	—	—	—	—	—	—	—	—	—
2.4	0	0.52	0.43	0.20	0.06	0.02	—	—	—	—	—	—	—	—	—
2.5	−0.05	0.50	0.45	0.22	0.07	0.02	0.01	—	—	—	—	—	—	—	—
3.0	−0.26	0.34	0.49	0.31	0.13	0.04	0.01	—	—	—	—	—	—	—	—
4.0	−0.40	−0.07	0.36	0.43	0.28	0.13	0.05	0.02	—	—	—	—	—	—	—
5.0	−0.18	−0.33	0.05	0.36	0.39	0.26	0.13	0.05	0.02	—	—	—	—	—	—
5.45	0	−0.34	−0.12	0.26	0.40	0.32	0.19	0.09	0.03	0.01	—	—	—	—	—
6.0	0.15	−0.28	−0.24	0.11	0.36	0.36	0.25	0.13	0.06	0.02	—	—	—	—	—
7.0	0.30	0.00	−0.30	−0.17	0.16	0.35	0.34	0.23	0.13	0.06	0.02	—	—	—	—
8.0	0.17	0.23	−0.11	−0.29	−0.10	0.19	0.34	0.32	0.22	0.13	0.06	0.03	—	—	—
8.65	0	0.27	0.06	−0.24	−0.23	0.03	0.26	0.34	0.28	0.18	0.10	0.05	0.02	—	—
9.0	−0.09	0.25	0.14	−0.18	−0.27	−0.06	0.20	0.33	0.31	0.21	0.12	0.06	0.03	0.01	—
10.0	−0.25	0.05	0.25	0.06	−0.22	−0.23	−0.01	0.22	0.32	0.29	0.21	0.12	0.06	0.03	0.01

Ancho de banda → Bw = 2*Jn*Fm = 2*5*0,15/3 MHz = 0,5 Mhz

Bw > 0,2 Mhz per tant banda ampla

P5
Les màquines no estan relacionades entre elles PER AIXÒ * P(B) i no P(A|B)

Les dues funcionin
$P(A \cap B)$ = P(A) * P(B)

Almenys una funcioni
P(AUB) = P(A) + P(B) - $P(A \cap B)$

beta= ΔFmax/Fm

FC LAB 13/05

1. Factor 20 — 30000 mostres

$\frac{x}{30000} = 20 \Rightarrow 20 \cdot 30000 = 600000$ mostres

2. Portadora $F_c = 10$ MHz $A_c = 0,07$ V — 40% — AM Pract 2
   moduladora $F_m = 150$ kHz $A_m = 1$ V — $m = 0,4$ — Script

   P? $R = 50\Omega$ en aquest cas = -30 dBm aprox → Mitjana y lateral

3. $F_c = 5$ MHz — FM
   $F_m = 350$ kHz
   $\beta \rightarrow$ Funció índex $\beta = 0$
   ↳ $J_3(\beta) = 0$

   Quan $F_{(H\bar{z})} = 0$?

   Quan $F_c \pm 3F_m = 0$ — Senyal F_m (6,05 MHz) = 0 V
   $5M \pm 3 \cdot 350k = 0$ — Senyal F_m (3,95 MHz) = 0 V

   6,05 MHz / 3,95 MHz

4. $F_a = 2,5$ kHz — 3 períodes — T_{OBS}?

   $T = \frac{1}{F_a} = 0,4$ ms → $T_{OBS} = 3T = 3 \cdot 0,4 = 1,2$ ms

5. $F_a = 2$ kHz $A_a = 4$ V — $F_S = 2,5$ M mostres/s
   30000 mostres
   afegim 120000 zeros al zero padding

   $\Delta f = \frac{F_S}{N_{total}}$

   $\Delta f = \frac{2,5M}{120000 + 30000} = \frac{2,5M}{150k} = 50$ Hz

6. Senyal quadrat $T_{OBS} = 126$ ms → 10,5 períodes F_a?

   $\frac{126\text{ms}}{x} = \frac{10,5}{1} \Rightarrow \frac{126m}{10,5} = 12\text{ms} = T \Rightarrow \frac{1}{12} = 83,3$ Hz

8. 'M1 i M2 INDEPE — cap funcioni? — Script L.
   0,3 − 0,5 → Funcions — sim = 10000
   ↓
   0,7 0,5 → No funcions → aplicar al script
   0,7 · 0,5 = 0,35 ≈ 0,34 del script

14/05

1. BLS $S_{modulada}$ $x = \cos(2 * pi * 160\,kHz * t)$

$A_c = 0{,}25V$ $FC = 8\,MHz$ Valor max de la sinc?

$V_{max} = 0{,}125\,V$

BLS
SCRIPT
Pract 3

2. FM modulada $x(t) = \cos(2600\pi t)$

$c(t) = \cos(7 \cdot 10^6 \pi t)$

Script
FM
Pract 4

$f_c - 4 \cdot f_m = 0 \rightarrow J_4(\beta) = 0$ cuan $\beta = 1{,}5$

$\beta = 8 \Rightarrow$ β entre 5 i 9

$f_c = \frac{7 \cdot 10^6 \pi}{2\pi} = 3{,}5\,MHz$

$f_m = \frac{2600\pi}{2\pi} = 1300\,Hz = 1{,}3\,kHz$

3. $\beta = 8 \rightarrow$ mira taula 14 j $2 \cdot 11 \cdot 1{,}3 = 28{,}6\,kHz$

$Bw = 2 \cdot 4 \cdot f_m = 2 \cdot 4 \cdot 1{,}3\,kHz = 10{,}4\,kHz$

4. $f_s = 3 \cdot 10^5$ mostres/s i 12000 mostres

$DIM = FS \cdot T_{obs} \Rightarrow T_{obs} = \frac{DIM}{FS} = \frac{3 \cdot 10^5}{12000} =$

7. FM $g(t) = 0{,}5 \cos(10^6 \pi t + 1{,}5 \sin(6000\pi t))$ ΔF? t_{mb}

$\beta = 1{,}5 \rightarrow$ Bessel $\rightarrow$ f_c i J_4

$\Delta F = k_f \cdot A_m$

$f_c = \frac{10^6 \pi}{2\pi} = 500\,kHz$ $f_m = \frac{6000}{2\pi} = 3\,kHz$

$\beta = \frac{\Delta f_{max}}{f_m} = 1{,}5 \cdot 3\,kHz = 4{,}5\,kHz$ (β ↑, f_m ↑)

8. Bw del 7?

4 J's $\rightarrow Bw = 2 \cdot 4 \cdot f_m = 2 \cdot 4 \cdot 3 = 24\,kHz$